

# CHAYAN AGGARWAL

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## EDUCATION

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### Indian Institute of Technology Roorkee

Roorkee, India

BS-MS in Mathematics and Computing

Aug 2025 – Present

## PROJECTS & EXPERIENCE

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### DeObfusca-AI - An ML model for deobfuscation

- Architected a machine learning pipeline utilizing Graph Neural Networks (GNNs) and Transformers to reverse-engineer control-flow obfuscated binaries.
- Integrated advanced neuro-symbolic AI to enhance the logical accuracy and reliability of the code deobfuscation process.
- Formulated a generalizable extension for analyzing VM-obfuscated binaries by transitioning the pipeline architecture from static Control Flow Graphs to dynamic Execution Trace Graphs.

### CHADChat - Steganography based decentralized messaging app

- Engineered a decentralized messaging application utilizing IPFS and Lighthouse for secure, immutable backend storage, coupled with a responsive dual-mode frontend for intuitive steganographic communication.
- Developed an on-device AI steganography model trained within a Generative Adversarial Network (GAN) framework, significantly enhancing robustness against adversarial perturbations and statistical steganalysis.
- Hardened the steganographic pipeline by integrating public-key cryptography, designed to mitigate brute-force and algebraic cryptanalysis from both classical and post-quantum vectors.
- Benchmarked the AI embedding process against standard security attack models, demonstrating superior payload concealment and structural integrity compared to traditional baseline algorithms.

### Kaggle Video Anomaly Detection

- Engineered a custom ConvLSTM Autoencoder model to identify irregular patterns in complex video datasets.
- Refined the model architecture for optimal feature extraction and developed streamlined inference scripts.
- Secured 6th position on the kaggle competition in which i submitted the model

### Custom Programming Language Interpreter

- Engineered a fully functional, dynamically typed programming language interpreter from scratch in Java - Heavily Inspired by Crafting interpreters book
- Implemented a robust tree-walk evaluation mechanism supporting complex language features including lexical scoping, closures, and object-oriented paradigms.
- Architected a native binding interface to seamlessly inject Java methods into the evaluation environment, constructing a standard library with mathematical operations and File I/O capabilities
- Designed static analysis passes for variable resolution and environment management to ensure correct binding of local and global scopes.

## SKILLS

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- **Programming Languages:** C/C++, Python, Rust, JavaScript, Java, Solidity, Assembly
- **Libraries/Frameworks:** Pytorch, NumPy, React.js, Node.js, PyCryptodome, pwntools
- **Knowledge:** Machine Learning, Low-Level Programming (OS, Compilers), Reverse Engineering, Binary Exploitation(pwn), Data Structures & Algorithms, Deep Learning, Git, Version Control, Proficient in Cryptographic Algorithms

## ACADEMIC ACHIEVEMENTS

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- Qualified Jee Advanced with a All India Rank of **1408**
- Core Member of the **Data Science Group** at IIT Roorkee
- Core Member of **MDG Space** at IIT Roorkee
- Secured a **Top 100** global rank in Backdoor CTF organised by InfoSec IITR - in a team of 2 where maximum size of a team was 4